# Kavennesh Balachandar

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## SUMMARY

Cybersecurity Engineer specializing in application security, penetration testing, vulnerability assessment, network defense, and SIEM detection engineering. Experienced in OWASP Top 10 remediation, firewall policy enforcement, automated incident response, and threat detection across Linux and cloud environments. Combines hands-on security engineering with vulnerability research, MCP server security scanning, and peer-reviewed research, supported by CompTIA Security+, CRTA, and eJPT certifications.

## EXPERIENCE

**Cybersecurity Analyst (Intern) - Application Security, Community Dreams Foundation | Remote** *Aug 2026 – Present*

- Remediated OWASP Top 10 vulnerabilities in a production HR platform, tracing injection, access-control, and authentication flaws through verification of corrective fixes.
- Implemented authentication, role-based access controls, and API authorization with input validation and secret rotation, strengthening protection across platform user roles.
- Hardened Supabase, PostgreSQL row-level security, and cloud settings, establishing audit-ready logs for authentication events and sensitive data-access activity across the platform.

**Graduate Teaching Assistant, Florida International University** *Aug 2025 – Apr 2026*

- Strengthened SQL Server role permissions and stored-procedure access across database administration labs, supporting secure configuration practices for more than 60 students.
- Rebuilt lab setup guides and troubleshooting workflows, reducing recurring student errors and improving practical assessment performance by 20% across database administration sessions.
- Automated T-SQL security configuration checks and audit-log analysis, then published administration guides adopted as the standard reference by 60+ students.

**Graduate Teaching Assistant, Florida International University** *Jan 2025 – Aug 2025*

- Designed SIEM and log-analysis modules covering centralized logging, alert correlation, detection rules, and MITRE ATT&CK mapping for 60+ cybersecurity students.
- Corrected iptables, firewalld, and Snort misconfigurations across Linux firewall and IDS lab hosts; revised primers to increase completion rates by 22%.
- Automated lab-submission processing with custom scripts while supporting Wireshark captures and IDS rule-writing exercises, reducing grading turnaround time by 40%.

**Network Security Analyst, Inter Deccan Solutions | Chennai, India** *Apr 2022 – Mar 2024*

- Reconfigured Cisco Secure Firewall 3100 zone policies, NAT, and ACLs across three enterprise sites, reducing unauthorized access by 40% within six weeks.
- Investigated production network misconfigurations using Nmap, Wireshark, and tcpdump, documenting vulnerability findings for monthly risk reviews and remediation prioritization discussions.
- Automated alert-triggered IP blocking, session termination, patch verification, and 12+ Linux tasks, cutting MTTR by 35% and saving 6–8 hours weekly.

## PROJECTS

**VPN Blocker: Detection & Prevention of VPN Tunneling - Peer-Reviewed** *2024*

- Developed VPN-tunneling detection using flow metadata, handshake timing, and packet-size entropy, achieving 90–95% accuracy without decrypting traffic across encrypted VPN protocols.
- Evaluated VPN detection with ACL enforcement, reducing policy violations by 50–60% while preserving payload confidentiality across OpenVPN, WireGuard, and HTTPS-obfuscated traffic.

**MCPScan - MCP Server Security Scanner** *Jul 2026 – Aug 2026*

- Built nine MCP security rules combining AST taint analysis and Docker-isolated probes, scanning 77 servers and 1,289 tool definitions with 822 tests.
- Shipped 9 rules pairing AST taint analysis with Docker-sandboxed probing across 77 servers and 1,289 tool definitions, backed by 822 tests across Python 3.11–3.13

**Cyber Range Lab - Penetration Testing Environment** *Apr 2026 – May 2026*

- Created a Docker-isolated penetration-testing range with DVWA and OWASP Juice Shop behind UFW segmentation, enabling repeatable exploitation testing without exposing production systems.
- Tested SQL injection, credential attacks, cross-site scripting, and command injection in isolated targets, achieving 95%+ success and cutting setup from 30 minutes to five.

**Home Networking Lab - DPI Detection & ELK SIEM** *May 2026 – Jul 2025*

- Classified more than 100 encrypted OpenVPN, WireGuard, and IPSec sessions using traffic fingerprints, addressing tunnel-detection gaps left by port-based network rules.
- Deployed ELK ingestion, Kibana dashboards, and anomaly alerts for encrypted-session monitoring, achieving 95% classification accuracy and cutting response time from 30 minutes to eight.

## CERTIFICATIONS AND AWARDS

**Certifications:** OSCP (Current), CompTIA Security+ (SY0-701), Certified Red Team Analyst (CRTA), eJPT, INE Certified Cloud Associate (ICCA), CCSP-AWS (Merit), CNSP (Merit)

**Awards & CTF:** Amazon AppSec CTF - NAMER Region, Eraksha Cybersecurity Competition - 1st Place National, Kavach Hackathon - National Qualifier

## EDUCATION

**Florida International University - Master of Science, Cyber Security** *Aug 2024 – Apr 2026*

**Anna University - Bachelor of Engineering, Computer Science** *Nov 2020 – Mar 2024*

## SKILLS

**Application & Offensive Security:** OWASP Top 10, Penetration Testing, Vulnerability Assessment, Burp Suite, Metasploit, SQL Injection, XSS, IDOR, OAuth Exploitation, AST Taint Analysis

**Network & Infrastructure Security:** Cisco Secure Firewall 3100, ACLs, NAT, iptables, firewalld, Snort, Suricata, TCP/IP, Deep Packet Inspection, Nmap, Wireshark, tcpdump

**SIEM & Threat Detection:** ELK Stack, Kibana, Log Analysis, Alert Correlation, Detection Engineering, Incident Response, MITRE ATT&CK

**Application & Cloud Security:** Supabase, PostgreSQL, Row-Level Security (RLS), API Authorization, Role-Based Access Control (RBAC), Authentication Security, Secrets Management, AWS IAM, CloudTrail, GuardDuty

**Security Automation & Development:** Python, Bash, PowerShell, SQL, T-SQL, Docker, CI/CD Security Gating, Security Configuration Automation

**AI Security & Research:** Model Context Protocol (MCP) Security, MCP Server Scanning, Tool Poisoning Detection, AST Taint Analysis, VPN Tunnel Detection

**Database & Systems Security:** Microsoft SQL Server, Stored-Procedure Permissions, Database Auditing, Linux Security Hardening